

HARSOVA, Romania

WMO: 154060

Lat: 44.6917N

Lon: 27.9636E

Elev: 39

StdP: 100.86

Time Zone: 2.00 (EUE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.8	-13.9	1.1	-10.1	-11.3	1.4	-7.4	17.6	-8.1	15.5	-3.5	3.5	0	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	33.5	20.9	32.2	20.6	30.8	20.3	23.2	29.7	22.2	28.7	21.5	27.9	3.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.5	16.2	25.3	20.5	15.2	24.3	19.6	14.4	23.5	69.0	29.4	65.6	28.4	62.7	28.1	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	9.6	7.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-1.0	2.5	6.3	11.2	17.4	21.9	23.4	23.3	18.5	12.0	6.9	1.7
	DBStd	9.34	4.58	5.01	4.70	4.12	3.14	2.81	2.53	2.87	3.92	4.19	4.90	4.35
	HDD10.0	1127	342	213	133	35	1	0	0	0	28	117	258	
	HDD18.3	2797	600	443	374	215	54	4	0	2	47	199	343	515
	CDD10.0	1876	0	3	17	71	230	358	415	411	255	90	24	2
	CDD18.3	505	0	0	0	2	24	113	157	154	52	3	0	0
	CDH23.3	4857	0	0	3	33	253	1021	1512	1543	468	23	1	0
	CDH26.7	1884	0	0	0	4	51	361	620	686	161	2	0	0
Wind	WSAvg	3.0	3.3	3.5	3.6	3.4	2.8	2.6	2.5	2.6	2.6	2.7	2.9	3.2
Precipitation	PrecAvg	469	30	20	31	34	49	58	53	37	47	35	34	37
	PrecMax	668	88	68	76	80	140	104	159	117	128	67	80	101
	PrecMin	264	2	2	1	2	9	16	10	5	4	2	1	4
	PrecStd	114	22	15	17	18	32	25	38	29	33	19	20	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	18.0	22.3	26.7	30.3	34.1	35.5	35.5	33.2	26.1	21.6	14.0
		MCWB	8.5	12.1	12.3	17.2	19.0	22.6	23.0	21.0	18.9	17.7	15.6	10.9
	2%	DB	8.7	14.0	18.9	23.3	27.9	32.0	33.2	33.6	30.3	23.5	18.1	11.7
		MCWB	7.0	10.3	11.2	14.9	18.5	21.4	21.1	20.5	18.3	15.9	13.8	9.4
	5%	DB	6.6	11.4	16.1	21.1	26.0	30.4	31.8	32.2	28.1	21.0	15.5	9.4
		MCWB	5.5	8.4	9.9	14.0	17.5	20.8	20.8	20.3	18.0	15.0	12.6	8.0
	10%	DB	4.9	8.9	13.3	18.9	24.3	28.7	30.2	30.6	25.7	18.7	13.5	7.1
		MCWB	4.1	7.1	8.6	12.5	16.8	20.1	20.4	19.9	17.4	13.9	11.5	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	12.9	13.4	18.0	20.7	24.1	25.6	24.3	21.6	18.9	16.2	12.0
		MCDB	10.7	17.3	20.6	25.7	27.4	31.7	30.0	30.9	27.3	24.5	20.2	13.9
	2%	WB	7.1	10.5	11.5	16.2	19.7	22.7	23.2	22.9	20.5	16.9	14.4	9.9
		MCDB	8.3	13.5	17.0	22.3	26.2	29.8	29.2	29.4	25.4	21.1	17.2	11.2
	5%	WB	5.8	8.6	10.3	14.6	18.5	21.8	22.3	21.8	19.6	15.9	13.2	8.0
		MCDB	6.8	10.8	15.0	19.9	24.1	28.5	28.3	28.6	24.7	20.2	15.6	9.7
	10%	WB	4.2	7.1	9.0	13.3	17.6	21.0	21.5	21.1	18.5	14.6	11.8	6.1
		MCDB	5.0	8.8	12.4	17.9	22.7	27.4	27.4	28.0	24.9	18.0	13.7	7.1
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.4	9.3	10.8	11.4	11.8	11.9	12.5	11.4	10.0	7.2	6.1
		MCDBR	8.0	11.2	14.2	14.6	14.1	14.1	14.1	15.0	14.5	13.3	10.1	8.8
		MCWBR	6.3	7.2	7.5	7.5	6.0	5.4	4.7	4.9	5.1	6.3	6.0	6.3
	5% WB	MCDBR	8.0	10.5	12.3	13.8	12.6	13.0	12.4	13.1	12.8	11.8	9.2	8.6
		MCWBR	6.4	7.2	7.1	7.4	6.1	5.3	5.1	5.0	5.3	6.3	5.5	6.3
Clear-Sky Solar Irradiance	taub	0.344	0.374	0.407	0.452	0.428	0.430	0.445	0.456	0.425	0.390	0.366	0.345	
	taud	2.293	2.224	2.179	2.067	2.221	2.249	2.217	2.174	2.229	2.307	2.322	2.317	
	Ebn at Noon	749	792	817	809	845	842	824	797	790	766	717	706	
	Edh at Noon	84	107	128	156	138	135	137	139	121	98	82	75	
All-Sky Solar Radiation	RadAvg	1.30	2.12	3.32	4.72	5.97	6.44	6.48	5.80	4.22	2.67	1.47	1.12	
	RadStd	0.19	0.30	0.34	0.47	0.39	0.40	0.37	0.33	0.32	0.27	0.18	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-122	+89	+57

Nomenclature: See separate page